

The Expected Hardy Envelope at the Lorentz Endpoint

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Source question and result

For a rearrangement-invariant quasi-Banach function space $Z \subset L^1(\mathbb{T})$, write

$$H[Z] = \{f \in Z : \widehat{f}(n) = 0 \text{ for every } n < 0\}$$

and denote Banach envelopes by a superscript $^\wedge$. Question 5.8 of Leśnik (Question 4 in the arXiv source) asks whether

$$H[Z]^\wedge = H[Z^\wedge] \tag{1}$$

for every separable rearrangement-invariant quasi-Banach function space $Z \subset L^1$. The source proves this when Z has nontrivial Boyd indices, using boundedness of the Riesz projection on Z .

We settle a genuine missing endpoint family. For $0 < q < 1$, let

$$\|f\|_{1,q} = \left(\int_0^1 [tf^*(t)]^q \frac{dt}{t} \right)^{1/q}, \quad Z = L^{1,q}(\mathbb{T}).$$

Here the upper Boyd index is 1, so the source lemma does not apply.

Theorem 1 (Lorentz endpoint). *For every $0 < q < 1$, the canonical identity on analytic polynomials extends to an isomorphism*

$$H[L^{1,q}(\mathbb{T})]^\wedge \simeq H^1(\mathbb{T}) = H[(L^{1,q}(\mathbb{T}))^\wedge]. \tag{2}$$

Thus the answer to (1) is affirmative for all $Z = L^{1,q}$, $0 < q < 1$.

Idea of the Proof

The lattice envelope of $L^{1,q}$ is L^1 . The missing point is that the analytic subspace has no larger envelope. Project a normalized real H^1 atom onto its nonnegative Fourier modes. An L^2 estimate near its support and cancellation away from the support give a two-piece decreasing rearrangement bound. This bound is uniformly integrable in the $L^{1,q}$ quasi-norm. A finite atomic decomposition of the real part of an analytic polynomial then bounds its analytic envelope norm by its H^1 norm. The opposite inequality is the elementary embedding $L^{1,q} \hookrightarrow L^1$.

To pass from polynomials to the full $H[L^{1,q}]$, we prove a density lemma which does not use the Riesz projection. Inner-outer factorization and an outer cutoff approximate any $H[Z]$ function by bounded analytic functions; radial approximation then gives analytic polynomials.

Two lemmas

Let P_+ denote the nonnegative-frequency projection when it is applied to a trigonometric polynomial or an L^2 function.

Lemma 1 (outer truncation gives polynomial density). *Let $Z \subset L^1(\mathbb{T})$ be a separable rearrangement-invariant quasi-Banach function space. Then the analytic polynomials $\mathcal{P}_A$ are dense in $H[Z]$.*

Proof. Separability of Z is equivalent to order continuity of its quasi-norm. Take nonzero $f \in H[Z] \subset H^1$ and write its inner-outer factorization as $f = IO$. Put

$$w_N = \min\{1, N/|O|\}, \quad u_N = \log w_N \leq 0,$$

and let h_N be the outer function with boundary modulus w_N , normalized in the usual way. Since

$$\|u_N\|_1 = \int_{\{|O|>N\}} \log(|O|/N) dm \longrightarrow 0,$$

weak type $(1, 1)$ of harmonic conjugation gives $\tilde{u}_N \rightarrow 0$ in measure. Hence $h_N \rightarrow 1$ in measure, while $|h_N| \leq 1$. The functions $f_N = fh_N$ are bounded analytic functions, because $|f_N| = \min\{|f|, N\}$ almost everywhere. Moreover,

$$|f_N - f| \leq 2|f|,$$

so order-continuous dominated convergence gives $\|f_N - f\|_Z \rightarrow 0$.

For $g \in H^\infty$, the radial functions $g_r(z) = g(rz)$ converge almost everywhere to g and are dominated by $2\|g\|_\infty$ after subtraction. The same dominated-convergence argument gives $g_r \rightarrow g$ in Z . For fixed $r < 1$, the Taylor polynomials of g_r converge uniformly on $\mathbb{T}$, hence in Z . Combining the two approximations proves the claim. $\square$

Lemma 2 (uniform analytic atom estimate). *Fix $0 < q < 1$. There is $C_q < \infty$ such that*

$$\|P_+a\|_{1,q} \leq C_q \tag{3}$$

for every real $(1, 2)$ Hardy atom a on $\mathbb{T}$: a is supported on an arc I of length δ , has integral zero, and $\|a\|_2 \leq \delta^{-1/2}$. The exceptional constant atom obeys the same conclusion.

Proof. Write $b = P_+a$. The L^2 boundedness of P_+ and Chebyshev's inequality give

$$b^*(t) \leq C(\delta t)^{-1/2} \quad (0 < t \leq C\delta). \tag{4}$$

Outside the doubled arc, the atom itself vanishes. Subtracting the conjugate-kernel value at the center c_I and using $\int a = 0$ gives

$$|b(e^{i\theta})| \leq \frac{C}{\text{dist}(\theta, I)^2} \int_I |s - c_I| |a(s)| ds \leq \frac{C\delta}{\text{dist}(\theta, I)^2},$$

because $\|a\|_1 \leq 1$. Consequently,

$$b^*(t) \leq C\delta t^{-2} \quad (C\delta \leq t \leq 1). \tag{5}$$

Using (4) and (5),

$$\|b\|_{1,q}^q \leq C^q \int_0^{C\delta} (t/\delta)^{q/2} \frac{dt}{t} + C^q \int_{C\delta}^1 (\delta/t)^q \frac{dt}{t} \leq C_q^q.$$

The constant atom is immediate. $\square$

Proof of Theorem 1

First, $L^{1,q} \hookrightarrow L^1$. One quick proof is to discretize the decreasing rearrangement on dyadic intervals: the L^1 norm is comparable to $\sum_k 2^{-k} f^*(2^{-k})$, whereas the q -th power of the Lorentz quasi-norm is comparable to the sum of the q -th powers of the same quantities. The inequality $\|x\|_{\ell^1} \leq \|x\|_{\ell^q}$ for $q < 1$ gives the embedding.

The Köthe associate of $L^{1,q}$ is L^∞ . One inclusion follows from the preceding embedding. For the converse, if g is essentially unbounded, testing on a normalized characteristic function of a set on which $|g|$ is arbitrarily large makes the associate norm infinite. Hence the Köthe bidual is L^1 . The standard envelope theorem for separable rearrangement-invariant quasi-Banach lattices contained in L^1 now gives

$$(L^{1,q})^\wedge = L^1 \quad (6)$$

with equivalent norms.

It remains to compare the two norms on $\mathcal{P}_A$. The embedding $L^{1,q} \hookrightarrow L^1$ implies, for every analytic polynomial p ,

$$\|p\|_1 \leq C_q \|p\|_{H[L^{1,q}]^\wedge}. \quad (7)$$

Indeed, apply the embedding to each term of an arbitrary finite decomposition of p in $H[L^{1,q}]$ and then take the infimum.

For the reverse inequality, put $c = p(0)$ and $u = \operatorname{Re}(p - c)$. This is a smooth real H^1 function of mean zero, and its real Hardy norm satisfies

$$\|u\|_{H_{\text{real}}^1} \leq C(\|u\|_1 + \|\tilde{u}\|_1) \leq C\|p\|_1.$$

The finite atomic characterization of real H^1 with $(1, 2)$ atoms gives an exact finite decomposition

$$u = \sum_{j=1}^M \lambda_j a_j, \quad \sum_{j=1}^M |\lambda_j| \leq C\|p\|_1.$$

Since $p - c$ is analytic and has zero constant coefficient,

$$p = c + 2P_+ u = c + 2 \sum_{j=1}^M \lambda_j P_+ a_j.$$

Lemma 2, together with the fixed cost of the constant function, therefore yields

$$\|p\|_{H[L^{1,q}]^\wedge} \leq C_q \left(|c| + \sum_j |\lambda_j| \right) \leq C'_q \|p\|_1. \quad (8)$$

Thus the envelope norm and the H^1 norm are equivalent on $\mathcal{P}_A$. Lemma 1 says that $\mathcal{P}_A$ is dense in $H[L^{1,q}]$, while it is classically dense in H^1 . Taking completions in (7)–(8) and using (6) proves (2). $\square$

A reusable criterion and the remaining obstruction

The proof actually gives the following criterion. Suppose Z is separable, rearrangement invariant, $Z \subset L^1$, $Z^\wedge = L^1$, and

$$\sup\{\|P_+ a\|_Z : a \text{ is a real } (1, 2) \text{ } H^1 \text{ atom}\} < \infty.$$

Then the same finite-atomic argument and Lemma 1 prove $H[Z]^\wedge = H^1 = H[Z^\wedge]$.

This does not settle the source question in full. For a general Z , its Banach envelope need not be L^1 ; even when it is, the displayed uniform atom estimate is not a formal consequence of the envelope identity. A general proof would need an analytic decomposition adapted to the geometry of $Z^\wedge$, while a counterexample would need a continuous functional on $H[Z]$ that cannot extend to $H[Z^\wedge]$. Neither is produced here.

Eight focused upgrade routes were pursued: source-proof auditing, a direct projection removal, dual extension, Fejér/spectral convexification, endpoint counterexample tests, Lorentz atom estimates, outer-truncation density, and the general atom-criterion upgrade. The final obstruction is recorded in the accompanying attempt log.

Novelty check

Exact-question, arXiv-id, title, “Banach envelope Hardy Lorentz,” and “density analytic polynomials abstract Hardy spaces” searches were run in the local FA/Banach indexes and primary literature through 2026-08-11. The searches found the source question, classical envelope results for other Hardy–Lorentz regimes, and later polynomial-density results for separable *Banach* function spaces, but no answer to Question 5.8 and no statement of Theorem 1. The proof above is self-contained apart from standard inner–outer factorization, weak $(1, 1)$ conjugation, finite real- H^1 atomic decomposition, and the lattice envelope theorem.

References

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